

STAT293 - Examples

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2025-12-03

Step 1: Source

Step 2: Simulation Paramaters

```
set.seed(234)

# Dimensions
K <- 10      # Number of individuals
T_len <- 30   # Total number of time points
d <- 10      # Number of variables (Dimension)

# Model Structure
p <- 1       # VAR lag order
sparsity <- 0.05 # Sparsity level (proportion of non-zero coefficients)
heterogeneity <- "medium" # Level of heterogeneity ("low", "medium", "high")

# Data Generation & Noise
sigma_ep <- 0.1 # Standard deviation of noise
phi_max <- 0.9  # Upper bound for coefficient generation
phi_min <- 0.1  # Lower bound for coefficient generation

# Evaluation Settings
n_holdout <- 5 # Number of time points withheld for forecasting (holdout)

# Code below (do not modify)
# Heterogeneity
frac_common <- switch(heterogeneity,
                      "low"    = 2/3,
                      "medium" = 1/2,
                      "high"   = 1/3)

# total number of nonzero elements in each d x d transition matrix
n_nonzero_total <- max(1, round(sparsity * d * d))

# number of nonzero elements for common effects
n_common <- max(1, round(frac_common * n_nonzero_total))

# number of nonzero elements for unique (individual-specific) effects
n_unique <- n_nonzero_total - n_common
```

Step 3: Generate $\text{Var}(p)$ ILD for each individual

```
# Generate Data for All Subjects
# data_list[[k]] contains the observed data matrix for subject k
# individual_B is 4D array [d, d, p, K]

individual_B <- gen_individual_B(K, d, p, n_common, n_unique)

data_list <- lapply(1:K, function(k) {

  # Get coefficients for subject k
  # Dimensions: [d, d, p]
  subject_phi <- array(individual_B[, , k], dim = c(d, d, p))

  # Generate time series data
  # Returns a matrix of size [T_len, d]
  gen_VAR_p(subject_phi, T_len, p, d, sigma_ep)

})
```

Step 4: Split Training & Test Datasets

```
# Define the split index (The end of the training period)
split_idx <- T_len - n_holdout

# Create Training Data
train_data <- lapply(data_list, function(X) { X[1:split_idx, ] })

# Create Test Data (With Lookback)
# Include the last 'p' rows of the training set.
# The "initial conditions (lags)" for the first prediction in the test set.
test_data <- lapply(data_list, function(X) { X[(split_idx + 1 - p):T_len, ] })

# --- Note for Evaluation ---
# 'test_data' now has (n_holdout + p) rows. The first 'p' rows are history.
# Evaluate metrics only on the last 'n_holdout' rows (the actual test set).
```

Step 5: (a) Individual LASSO for $\text{VAR}(p)$

Fit Lasso $\text{VAR}(1)$ and $\text{VAR}(p)$ for each subject

```
# VAR(p)
if (p != 1){
  est_lasso_B_p <- lapply(train_data, function(data) {
    fit_lasso_var_p(data, p)
  })

  est_lasso_B_p <- array(unlist(est_lasso_B_p), dim = c(d, d, p, K))
}

# VAR(1)
```

```
est_lasso_B <- lapply(train_data, function(data) {
  fit_lasso_var_p(data, p = 1)
})

est_lasso_B <- array(unlist(est_lasso_B), dim = c(d, d, K))
```

Step 6: (b) Multi-VAR

Standard LASSO

```
#lambda_standard <- 10^seq(log10(0.01), log10(5), length.out = 30)

multivar_lasso_model <- constructModel(
  data = train_data,
  lag = 1,
  #lambda1 = lambda_standard,
  #lambda2 = lambda_standard,
  standardize = TRUE,
  canonical = FALSE,
  lassotype = "standard", # standard multi-VAR (not adaptive)
  cv = "blocked",
  nfolds = 10
)

multivar_lasso_res <- cv.multivar(multivar_lasso_model)

## |

est_multivar_lasso_B <- multivar_B(multivar_lasso_res$mats$total, d, K)
```

Adaptive

```
#lambda_adaptive <- 10^seq(log10(0.01), log10(5), length.out = 30)

multivar_adapt_model <- constructModel(
  data = train_data,
  lag = 1,
  #lambda1 = lambda_adaptive,
  #lambda2 = lambda_adaptive,
  standardize = TRUE,
  #weighttest = "ridge",
  canonical = FALSE,
  lassotype = "adaptive",
  cv = "blocked",
  nfolds = 10
)

multivar_adapt_res <- cv.multivar(multivar_adapt_model)

## |

est_multivar_adapt_B <- multivar_B(multivar_adapt_res$mats$total, d, K)
```

Step 7: Evaluation

Estimation Errors

```
B_true <- individual_B[ , , 1, ]

diffB_lasso <- diff_func(est_lasso_B, B_true)
diffB_multivar_lasso <- diff_func(est_multivar_lasso_B, B_true)
diffB_multivar_adapt <- diff_func(est_multivar_adapt_B, B_true)

diff_table <- data.frame(
  Method = c("LASSO (Lag 1)", "Multi-VAR (Standard)", "Multi-VAR (Adaptive)"),
  Error_Metric = c(diffB_lasso, diffB_multivar_lasso, diffB_multivar_adapt),
  stringsAsFactors = FALSE
)

if (p != 1) {
  diffB_lasso_p <- diff_func(est_lasso_B_p, individual_B)

  extra_row <- data.frame(
    Method = paste0("LASSO (Lag ", p, ")"),
    Error_Metric = diffB_lasso_p,
    stringsAsFactors = FALSE
  )

  diff_table <- rbind(diff_table, extra_row)
}

diff_table

##           Method Error_Metric
## 1      LASSO (Lag 1)    11.386448
## 2 Multi-VAR (Standard)     6.324307
## 3 Multi-VAR (Adaptive)    11.474808

kable(diff_table, digits = 3,
      caption = "Comparison of Estimation Error")
```

Table 1: Comparison of Estimation Error

Method	Error_Metric
LASSO (Lag 1)	11.386
Multi-VAR (Standard)	6.324
Multi-VAR (Adaptive)	11.475

Sensitivity and Specificity

```
# LASSO
metrics_lasso <- calc_metrics(
  B_true,
  B_est = est_lasso_B,
  output_type = "mean"
)
```

```

# multivar lasso
metrics_multivar_lasso <- calc_metrics(
  B_true,
  B_est = est_multivar_lasso_B,
  output_type = "mean"
)

# multivar adaptive
metrics_multivar_adapt <- calc_metrics(
  B_true,
  B_est = est_multivar_adapt_B,
  output_type = "mean"
)

# results
metrics_var1 <- data.frame(
  Method= c("LASSO (Lag 1)", "Multi-VAR (Standard)", "Multi-VAR (Adaptive)"),
  Sensitivity = c(metrics_lasso$mean_sens,
                  metrics_multivar_lasso$mean_sens,
                  metrics_multivar_adapt$mean_sens),
  Specificity = c(metrics_lasso$mean_spec,
                  metrics_multivar_lasso$mean_spec,
                  metrics_multivar_adapt$mean_spec)
)

ss_dat <- metrics_var1

if (p != 1) {
  metrics_lasso_p <- calc_metrics(
    B_true = individual_B,
    B_est = est_lasso_B_p,
    output_type = "mean"
  )

  metrics_varp <- data.frame(
    Method = "LASSO (Lag p)",
    Sensitivity = metrics_lasso_p$mean_sens,
    Specificity = metrics_lasso_p$mean_spec
  )

  ss_dat <- rbind(ss_dat, metrics_varp)
}

ss_dat

##           Method Sensitivity Specificity
## 1      LASSO (Lag 1)         0.70    0.8431579
## 2 Multi-VAR (Standard)         0.76    0.8400000
## 3 Multi-VAR (Adaptive)         0.20    0.9968421

kable(ss_dat, digits = 3,
      caption = "Comparison of Mean Sensitivity and Specificity")

```

Table 2: Comparison of Mean Sensitivity and Specificity

Method	Sensitivity	Specificity
LASSO (Lag 1)	0.70	0.843
Multi-VAR (Standard)	0.76	0.840
Multi-VAR (Adaptive)	0.20	0.997

Sensitivity and Specificity (per subject)

```

# LASSO
Smetrics_lasso <- calc_metrics(
  B_true,
  B_est = est_lasso_B,
  output_type = "subject"
)

# multivar_lasso
Smetrics_multivar_lasso <- calc_metrics(
  B_true,
  B_est = est_multivar_lasso_B,
  output_type = "subject"
)

# multivar_adaptive
Smetrics_multivar_adapt <- calc_metrics(
  B_true,
  B_est = est_multivar_adapt_B,
  output_type = "subject"
)

# results
ss_dat_subject <- data.frame(
  Method = rep(c("LASSO (Lag 1)", "Multi-VAR (Standard)", "Multi-VAR (Adaptive)"), each = K),
  Sensitivity = c(Smetrics_lasso$sensitivity,
                  Smetrics_multivar_lasso$sensitivity,
                  Smetrics_multivar_adapt$sensitivity),
  Specificity = c(Smetrics_lasso$specificity,
                  Smetrics_multivar_lasso$specificity,
                  Smetrics_multivar_adapt$specificity)
)

if (p != 1) {
  Smetrics_lasso_p <- calc_metrics(
    B_true = individual_B,
    B_est = est_lasso_B_p,
    output_type = "subject"
  )

  Smetrics_varp <- data.frame(
    Method = "LASSO (Lag p)",
    Sensitivity = Smetrics_lasso_p$sensitivity,
    Specificity = Smetrics_lasso_p$specificity
  )
}

```

```

)

ss_dat_subject <- rbind(ss_dat_subject, Smetrics_varp)
}

desired_order <- c("LASSO (Lag 1)",
                  "LASSO (Lag p)",
                  "Multi-VAR (Standard)",
                  "Multi-VAR (Adaptive)")

summary_table <- ss_dat_subject %>%
  group_by(Method) %>%
  summarise(
    # --- Sensitivity ---
    Sens_Mean = mean(Sensitivity, na.rm = TRUE),
    Sens_SD   = sd(Sensitivity, na.rm = TRUE),

    # --- Specificity ---
    Spec_Mean = mean(Specificity, na.rm = TRUE),
    Spec_SD   = sd(Specificity, na.rm = TRUE)
  ) %>%

  mutate(
    `Sensitivity Mean (SD)` = sprintf("%.3f (%.3f)", Sens_Mean, Sens_SD),
    `Specificity Mean (SD)` = sprintf("%.3f (%.3f)", Spec_Mean, Spec_SD),
    Method = factor(Method, levels = desired_order)
  ) %>%

  arrange(Method) %>%

  select(Method, `Sensitivity Mean (SD)`, `Specificity Mean (SD)`)

summary_table

## # A tibble: 3 x 3
##   Method                `Sensitivity Mean (SD)` `Specificity Mean (SD)`
##   <fct>                <chr>                  <chr>
## 1 LASSO (Lag 1)        0.700 (0.216)                0.843 (0.087)
## 2 Multi-VAR (Standard) 0.760 (0.207)                0.840 (0.039)
## 3 Multi-VAR (Adaptive) 0.200 (0.267)                0.997 (0.005)

kable(summary_table, caption = "Model Performance - Mean (SD) of Sensitivity and Specificity across Subj")

```

Table 3: Model Performance – Mean (SD) of Sensitivity and Specificity across Subjects

Method	Sensitivity Mean (SD)	Specificity Mean (SD)
LASSO (Lag 1)	0.700 (0.216)	0.843 (0.087)
Multi-VAR (Standard)	0.760 (0.207)	0.840 (0.039)
Multi-VAR (Adaptive)	0.200 (0.267)	0.997 (0.005)

RMSFE (Root Mean Square Forecast Error)

```
# LASSO
rmsfe_lasso <- get_rmsfe_p(est_lasso_B, test_data, K)

# Multi-VAR (Standard)
rmsfe_multivar_lasso <- get_rmsfe_p(est_multivar_lasso_B, test_data, K)

# Multi-VAR (Adaptive)
rmsfe_multivar_adapt <- get_rmsfe_p(est_multivar_adapt_B, test_data, K)

rmsfe_dat <- data.frame(
  Method = rep(c("LASSO (Lag 1)", "Multi-VAR (Standard)", "Multi-VAR (Adaptive)"), each = K),
  RMSFE = c(rmsfe_lasso, rmsfe_multivar_lasso, rmsfe_multivar_adapt))

if (p != 1) {
  rmsfe_lasso_p <- get_rmsfe_p(est_lasso_B_p, test_data, K)

  rmsfe_dat_varp <- data.frame(
    Method = "LASSO (Lag p)",
    RMSFE = rmsfe_lasso_p
  )

  rmsfe_dat <- rbind(rmsfe_dat, rmsfe_dat_varp)
}

# results
rmsfe_table <- rmsfe_dat %>%
  group_by(Method) %>%
  summarise(
    Mean = mean(RMSFE, na.rm = TRUE),
    SD = sd(RMSFE, na.rm = TRUE),
  ) %>%

  mutate(
    `RMSFE Mean (SD)` = sprintf("%.3f (%.3f)", Mean, SD),
    Method = factor(Method, levels = desired_order)
  ) %>%

  arrange(Method) %>%

  select(Method, `RMSFE Mean (SD)`)

rmsfe_table

## # A tibble: 3 x 2
##   Method          `RMSFE Mean (SD)`
##   <fct>          <chr>
## 1 LASSO (Lag 1)    0.106 (0.013)
## 2 Multi-VAR (Standard) 0.103 (0.012)
## 3 Multi-VAR (Adaptive) 0.105 (0.013)
```



```
kable(rmsfe_table, caption = "Model Performance - Mean (SD) of RMSFE across Subjects")
```

Table 4: Model Performance – Mean (SD) of RMSFE across Subjects

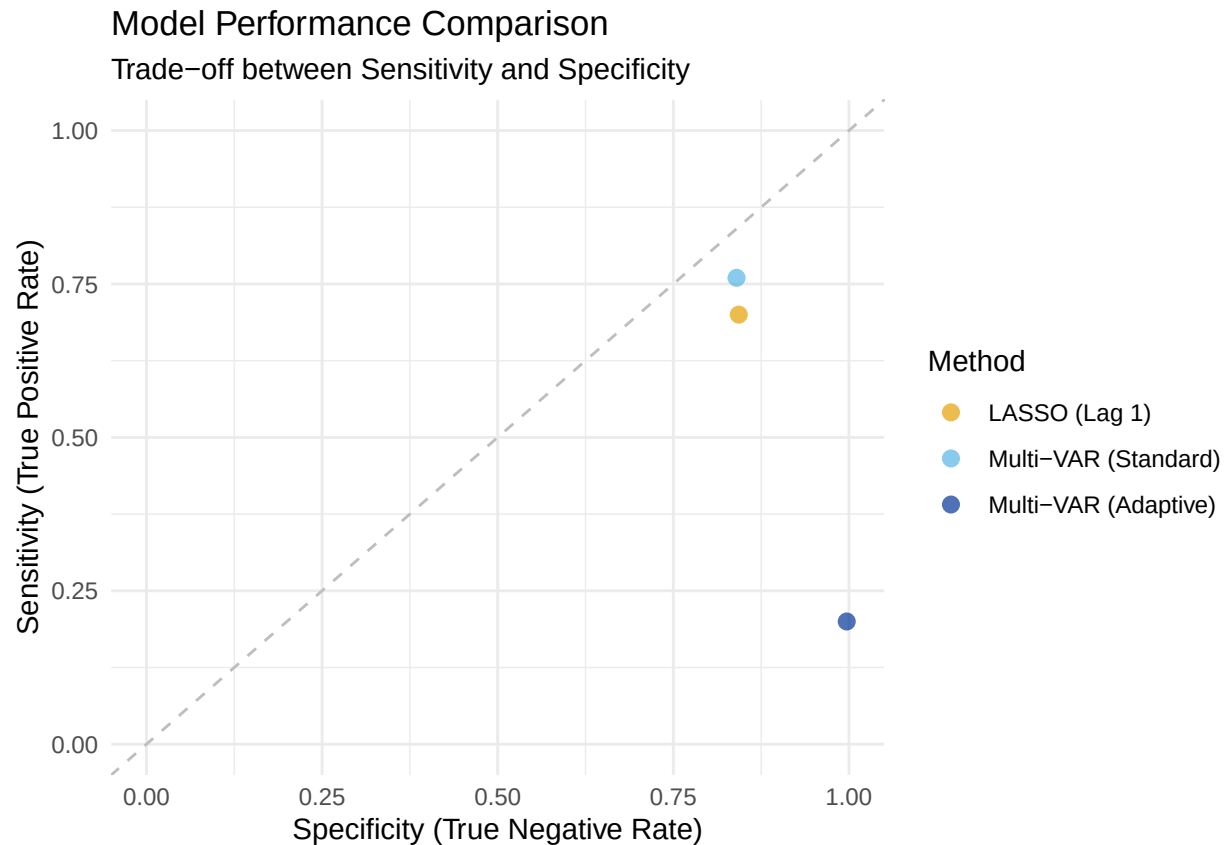
Method	RMSFE Mean (SD)
LASSO (Lag 1)	0.106 (0.013)
Multi-VAR (Standard)	0.103 (0.012)
Multi-VAR (Adaptive)	0.105 (0.013)

Step 8: Visualization

Sensitivity and Specificity (mean)

```
ss_dat$Method <- factor(ss_dat$Method, levels = desired_order)
color_lab <- c("LASSO (Lag 1)" = "#E69F00",
               "LASSO (Lag p)" = "#CC3300",
               "Multi-VAR (Standard)" = "#56B4E9",
               "Multi-VAR (Adaptive)" = "#003399")

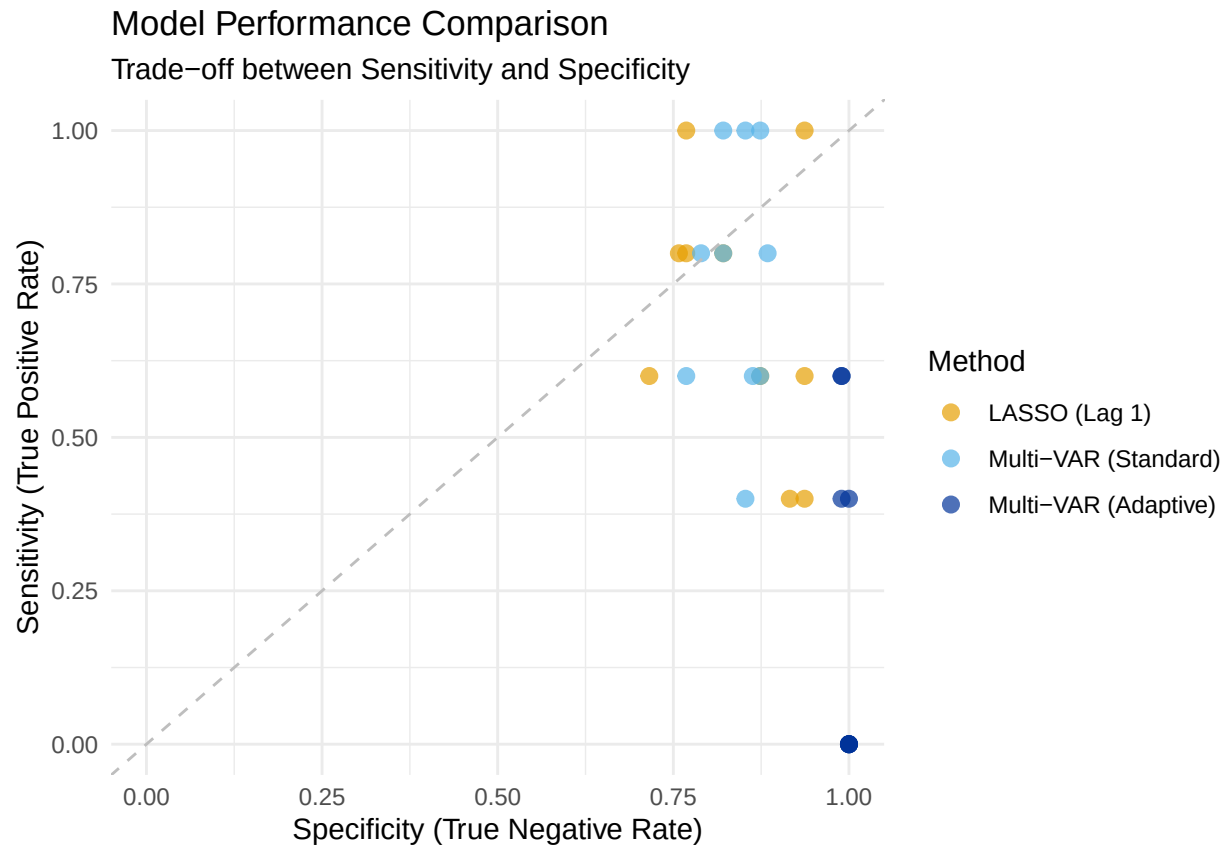
ggplot(ss_dat, aes(x = Specificity, y = Sensitivity, color = Method)) +
  geom_point(size = 2.5, alpha = 0.7) +
  xlim(0, 1) +
  ylim(0, 1) +
  geom_abline(intercept = 0, slope = 1, linetype = "dashed", color = "gray") +
  labs(
    title = "Model Performance Comparison",
    subtitle = "Trade-off between Sensitivity and Specificity",
    x = "Specificity (True Negative Rate)",
    y = "Sensitivity (True Positive Rate)"
  ) +
  theme_minimal() +
  scale_color_manual(values = color_lab)
```



Sensitivity and Specificity (per subject)

```
ss_dat_subject$Method <- factor(ss_dat_subject$Method, levels = desired_order)

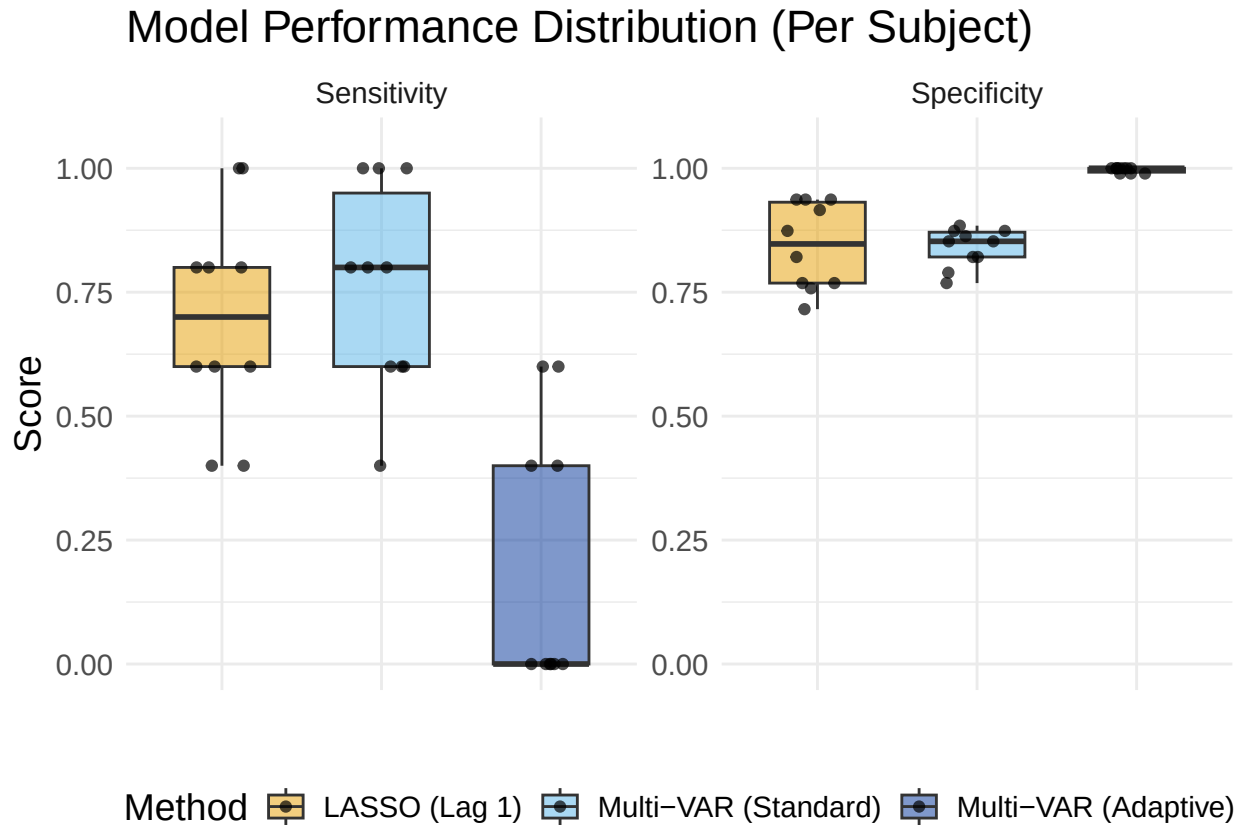
# scatter plot
ggplot(ss_dat_subject, aes(x = Specificity, y = Sensitivity, color = Method)) +
  geom_point(size = 2.5, alpha = 0.7) +
  xlim(0, 1) +
  ylim(0, 1) +
  geom_abline(intercept = 0, slope = 1, linetype = "dashed", color = "gray") +
  labs(
    title = "Model Performance Comparison",
    subtitle = "Trade-off between Sensitivity and Specificity",
    x = "Specificity (True Negative Rate)",
    y = "Sensitivity (True Positive Rate)"
  ) +
  theme_minimal() +
  scale_color_manual(values = color_lab)
```



```
# boxplot
ss_dat_subject_long <- melt(ss_dat_subject,
  id.vars = c("Method"),
  measure.vars = c("Sensitivity", "Specificity"),
  variable.name = "Metric",
  value.name = "Score")
ss_dat_subject_long$Method <- factor(ss_dat_subject_long$Method, levels = desired_order)

ggplot(ss_dat_subject_long, aes(x = Method, y = Score, fill = Method)) +
  geom_boxplot(alpha = 0.5, outlier.shape = NA, width = 0.6) +
  geom_jitter(width = 0.2, height = 0, size = 1.5, alpha = 0.7) +
  facet_wrap(~Metric, scales = "free_y") +
  scale_fill_manual(values = color_lab) +
  labs(
    title = "Model Performance Distribution (Per Subject)",
    #subtitle = "Boxplots show variability across 10 subjects",
    y = "Score",
    x = ""
  ) +
  ylim(0, 1.05) +
  theme_minimal() +
  theme(
    axis.text.x = element_blank(),
    axis.ticks.x = element_blank(),
    legend.position = "bottom",
    text = element_text(size = 14)
```

)



RMSFE (per subject)

```
# boxplot
rmsfe_dat$Method <- factor(rmsfe_dat$Method, levels = desired_order)

ggplot(rmsfe_dat, aes(x = Method, y = RMSFE, fill = Method)) +
  geom_boxplot(alpha = 0.5, outlier.shape = NA, width = 0.6) +
  geom_jitter(width = 0.2, height = 0, size = 1.5, alpha = 0.7) +
  scale_fill_manual(values = color_lab) +
  expand_limits(y = 0) +

  labs(
    title = "Forecast Error Distribution (RMSFE per Subject)",
    #subtitle = "Lower is Better; Boxplots show variability across subjects",
    y = "RMSFE",
    x = ""
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_blank(),
    axis.ticks.x = element_blank(),
    legend.position = "bottom",
    text = element_text(size = 14)
```

)

